05. Number Logic

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Spotting the rule in a number pair:

5 and 4 go together to make 9. 7 and 6 go together to make 13. Using the same rule, what do 8 and 10 make?

Answer: The rule is: add the two numbers together (5+4=9, 7+6=13), so 8+10 = **18**.

Example 2 — Odd one out:

Four of these five numbers share a property, but one does not belong: 16, 24, 32, 36, 40. Which number is the odd one out, and what property do the other four share?

Answer: **36** is the odd one out — the other four (16, 24, 32, 40) are all multiples of 8, but 36 is not.

Example 3 — Multi-step age problem:

Sam is 7 years old. His sister Ella is 3 years older than Sam. Their cousin Raj is twice as old as Ella. How old is Raj?

Answer: Ella = 7+3 = 10. Raj = 2×10 = **20 years old**.

Example 4 — Number grid:

In this number grid, each row follows the same rule. Row 1: 2, 4, 6. Row 2: 5, 3, 8. Row 3: 7, 1, ?. What is the missing number?

Answer: Each row follows the same rule: add the first two numbers to get the third (2+4=6, 5+3=8), so row 3: 7+1 = **8**.

Example 5 — Comparison logic (select-all):

Leo is taller than Mia. Mia is taller than Zara. Select ALL statements that must be true. A) Leo is taller than Zara B) Zara is the shortest of the three C) Mia is taller than Leo D) Leo is the tallest of the three

Answer: Since Leo > Mia > Zara, Leo is taller than Zara (A), Zara is the shortest (B) and Leo is the tallest (D) must all be true. C reverses a true comparison, so it is false. Answer: **A, B and D**.

Foundation

15 questions

1. 4 and 3 go together to make 7. 8 and 5 go together to make 13. Using the same rule, what do 6 and 9 make?

2. Four of these five numbers share a property, but one does not belong: 6, 12, 18, 25, 24. Which number is the odd one out, and what property do the other four share?

3. Leo is 5 years old. Mia is twice as old as Leo. Noah is 2 years older than Mia. How old is Noah?

4. 3 and 4 go together to make 7. 5 and 8 go together to make 13. Using the same rule, what do 6 and 7 make? A) 12 B) 13 C) 14 D) 15 E) 16

5. Four of these five numbers share a property, but one does not belong: 14, 21, 28, 33, 35. Which number is the odd one out, and what property do the other four share?

6. Ella scored 7 points in a quiz. Omar scored 5 more points than Ella. How many points did Omar score?

7. In this number grid, each row follows the same rule. Row 1: 2, 5, 7. Row 2: 4, 3, 7. Row 3: 6, 1, ?. What is the missing number?

8. 5 and 5 go together to make 10. 6 and 7 go together to make 13. Using the same rule, what do 9 and 8 make?

9. Zara is older than Leo. Leo is older than Finn. Select ALL statements that must be true. A) Zara is older than Finn B) Finn is the youngest of the three C) Leo is older than Zara D) Zara is the oldest of the three E) Finn is older than Leo

10. Four of these five numbers share a hidden property; one does not. Which is the odd one out? A) 15 B) 25 C) 35 D) 42 E) 55

11. Noah has 4 pencils. Zara has three times as many pencils as Noah. How many pencils does Zara have?

12. 9 and 3 go together to make 12. 7 and 8 go together to make 15. Using the same rule, what do 11 and 4 make?

13. Finn is 6 years old. Grace is 3 years older than Finn. Leo is 5 years younger than Grace. How old is Leo? A) 2 B) 3 C) 4 D) 5 E) 6

14. Four of these five numbers share a property, but one does not belong: 8, 12, 16, 20, 22. Which number is the odd one out, and what property do the other four share?

15. 6 and 6 go together to make 12. 9 and 2 go together to make 11. Using the same rule, what do 10 and 7 make? A) 15 B) 16 C) 17 D) 18 E) 19

Standard

20 questions

1. Four of these five numbers share a property, but one does not belong: 4, 9, 25, 30, 49. Which number is the odd one out, and what property do the other four share?

2. 9 and 2 go together to make 7. 12 and 5 go together to make 7. Using the same rule, what do 15 and 6 make?

3. Kira has 6 stickers. Leo has 4 more stickers than Kira. Priya has twice as many stickers as Leo. How many stickers does Priya have?

4. 3 and 2 go together to make 8. 5 and 1 go together to make 11. Using the same rule, what do 6 and 2 make? A) 12 B) 13 C) 14 D) 15 E) 16

5. Four of these five numbers share a property, but one does not belong: 15, 20, 24, 30, 35. Which number is the odd one out, and what property do the other four share?

6. Omar scored 65 in a test. Priya scored 15 more than Omar. Dev scored half of Priya's score. How many marks did Dev score?

7. In this number grid, each row follows the same rule. Row 1: 2, 3, 7. Row 2: 4, 1, 9. Row 3: 5, 2, ?. What is the missing number?

8. 6 and 1 go together to make 5. 9 and 4 go together to make 5. Using the same rule, what do 10 and 3 make?

9. 14 and 6 go together to make 8. 22 and 15 go together to make 7. Using the same rule, select ALL of these pairs that would make 9. A) 20 and 11 B) 16 and 7 C) 25 and 16 D) 12 and 4 E) 30 and 20

10. Four of these five numbers share a hidden property; one does not. Which is the odd one out? A) 18 B) 27 C) 33 D) 36 E) 44

11. Bake stall A sold 20 cupcakes. Bake stall B sold twice as many cupcakes as stall A. Bake stall C sold 12 fewer cupcakes than stall B. How many cupcakes did stall C sell?

12. 20 and 12 go together to make 8. 17 and 10 go together to make 7. Using the same rule, what do 25 and 14 make?

13. Priya is 9 years old. Her brother Dev is 3 years younger than Priya. Their cousin Mia is 6 years older than Priya. How old is Mia? A) 12 B) 13 C) 14 D) 15 E) 16

14. Four of these five numbers share a property, but one does not belong: 10, 15, 21, 25, 30. Which number is the odd one out, and what property do the other four share?

15. 16 and 9 go together to make 7. 23 and 8 go together to make 15. Using the same rule, what do 31 and 19 make? A) 10 B) 11 C) 12 D) 13 E) 14

16. Finn collected £25 for charity. Grace collected £15 more than Finn. Noah collected half as much as Grace. How much did Finn and Noah collect together?

17. In this number grid, each row follows the same rule. Row 1: 4, 2, 6. Row 2: 7, 3, 10. Row 3: 9, 5, ?. What is the missing number?

18. 5 and 2 go together to make 8. 7 and 3 go together to make 11. Using the same rule, what do 6 and 4 make?

19. Four of these five numbers share a property, but one does not belong: 13, 17, 19, 23, 27. Which number is the odd one out, and what property do the other four share?

20. 14 and 10 go together to make 2. 19 and 9 go together to make 5. Using the same rule, what do 22 and 14 make? A) 2 B) 3 C) 4 D) 5 E) 6

Hard

5 questions

1. Four of these five numbers share a property, but one does not belong: 18, 36, 45, 54, 61. Which number is the odd one out, and what property do the other four share?

2. 3 and 2 go together to make 10. 5 and 1 go together to make 12. Using the same rule, what do 4 and 3 make?

3. Priya collected £18 for charity. Her friend Zara collected twice as much as Priya. Their teacher collected £9 less than Zara. How much did Priya and the teacher collect together?

4. Mia is older than Noah. Noah is the same age as Omar. Omar is older than Priya. Select ALL statements that must be true. A) Mia is older than Priya B) Omar is older than Mia C) Noah is older than Priya D) Priya is older than Omar E) Mia is the oldest of the four

5. In this number grid, each row follows the same rule. Row 1: 3, 4, 11. Row 2: 5, 2, 13. Row 3: 6, 3, ?. What is the missing number?

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